

Report
Of
Machine Learning Intern

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
MCA



By

Roll Number of Student: 25SCS2040002855

Name of Student: Ayush Chaudhary

Batch: 2026-27

Chapter 1: CANDIDATE'S DECLARATION

I, Ayush Chaudhary do hereby declare that the internship report titled “**Data Analytics Intern**” has been completed by me to fulfil the requirement for the award of the degree of Master of Computer Application. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Ayush Chaudhary

Roll No.: 25SCS2040002855

Chapter 2: ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

Student Name: Ayush Chaudhary

Roll No.: 25SCS2040002855

Chapter No.	Chapter Name	Page Nos.
	Cover Page	----
1.	Candidate's Declaration	
2.	Acknowledgment	
3.	Internship Completion Certificate	
4.	Project Description 4 .1 Introduction 4.2 Organization Profile 4.3 Problem Statement 4.4 Project Objectives 4.5 Scope of the Project 4.6 Technologies and Tools Used 4.7 System Architecture 4.8 Methodology 4.9 Expected Outcomes 4.10 Certificates of completion and other communication mail proof must be attached	
5.	Bibliography/References	

Chapter 3: Internship Completion Certificate



Chapter 4: Project Description

4.1 Introduction

The Data Analytics Internship was structured as a comprehensive, hands-on, project-based professional program aimed at bridging academic concepts with real-world enterprise data analytical workflows. The internship provided practical exposure to the entire lifecycle of data analysis, emphasizing how raw, unstructured, or semi-structured transactional records are extracted, cleaned, modeled, and visualized to generate strategic business intelligence. Rather than focusing solely on isolated tools, the program emphasized practical problem-solving methodologies, key performance indicator (KPI) tracking, and actionable storytelling through visual media.

To build a robust foundation while working toward production-grade reporting, the project implementation was carried out across two distinct technical phases:

- **Phase 1: Exploratory Data Analysis & Programmatic Visualization (Python Ecosystem)** The foundational phase focused on programmatic data manipulation, numerical computation, and exploratory analysis. Using Python alongside core data science libraries—including Pandas, NumPy, Matplotlib, and Seaborn—the objective was to conduct rigorous data hygiene routines. This involved handling missing values, identifying outliers, performing categorical encoding, and computing key descriptive statistics. Programmatic visual plots, such as distribution charts, correlation heatmaps, and trend lines, were constructed to uncover hidden distributions, anomalies, and feature interactions within real-world datasets.
- **Phase 2: Enterprise Business Intelligence & Interactive Dashboarding (Microsoft Power BI)** Building upon the analytical findings of the programmatic phase, the advanced phase transitioned toward commercial business intelligence and executive reporting. Utilizing Microsoft Power BI, multi-dimensional transactional records from a commercial Superstore were processed into a coherent analytical data model. Advanced data transformation was conducted through Power Query, followed by the development of custom Data Analysis Expressions (DAX) to dynamically calculate critical business metrics, such as year-over-year revenue growth, regional profit margins, shipping performance, and category-wise sales distributions.

Through this structured progression from Python-based data cleaning to interactive Power BI reporting, the internship successfully simulated the functional responsibilities of an enterprise Data Analytics Intern, showcasing end-to-end data processing, interactive visual layout engineering, and data-driven business insight generation.

4.2 Organization Profile

Codec Technologies Pvt. Ltd. is an ISO 9001:2015 certified IT services, consultancy, and technical skill development enterprise. The company delivers enterprise-grade hosting solutions, cloud integrations, and custom software development across global markets while maintaining an active corporate training ecosystem recognized by national and international quality bodies, including the All-India Council for Technical Education (AICTE) and the International Certification Accreditation Council (ICAC).

Operating as both a tech-solutions provider and a structured skill-development platform, Codec Technologies bridges the gap between traditional academic curricula and modern industry expectations. The organization provides interns with exposure to live datasets, production-grade tools, and structured project frameworks across emerging tech domains such as Data Analytics, Machine Learning, Web Engineering, and Cloud Infrastructure.

The corporate culture at Codec Technologies emphasizes hands-on software development, rigorous documentation standards, continuous mentorship, and agile project execution. The firm's internal structure is broadly partitioned into four operational units:

- **Enterprise IT Consultancy & Digital Solutions:** Responsible for engineering scalable software systems, web applications, and automated workflow solutions for client organizations across varied business sectors.
- **Data Analytics & Business Intelligence Division:** Specializes in processing large volumes of operational data, designing relational and dimensional data models, and constructing interactive BI dashboards that empower enterprise leaders to make data-driven decisions.
- **Cloud Infrastructure & Managed Services:** Focuses on cloud migration strategies, high-performance hosting environments, database administration, and IT security protocols.
- **Corporate Training & Skill Acceleration Wing:** Manages AICTE-recognized internship programs, technical bootcamps, and industrial project-based modules designed to train emerging developers in enterprise-level tech stacks.

During the internship period, the Data Analytics & Business Intelligence Division served as the primary reporting unit. Working under the guidance of senior technical leads, interns executed end-to-end data pipelines—ranging from initial programmatic data manipulation in Python to executive dashboard design in Microsoft Power BI—ensuring adherence to modern data architecture and corporate analytics standards.

4.3 Problem Statement

Modern retail and commercial enterprises generate massive volumes of transactional data across various geographical regions, product categories, customer segments, and order fulfillment channels on a daily basis. However, organizations often struggle to extract timely, actionable insights from this operational data due to two primary technical bottlenecks:

- **Unstructured & Dirty Raw Data:** Raw transactional datasets frequently suffer from missing attributes, inconsistent data types, duplicate records, and unstructured formats. Without robust data cleaning and exploratory analysis, attempting to generate business metrics yields misleading conclusions, unaddressed anomalies, and flawed forecasting.
- **Static & Fragmented Business Reporting:** Legacy reporting methods rely heavily on static spreadsheets and tabular outputs. These formats fail to provide interactive visibility into critical Key Performance Indicators (KPIs)—such as net profit margins, order fulfillment lag times, regional sales performance, and customer segment profitability. Enterprise decision-makers are left unable to perform real-time slice-and-dice operations, isolate underperforming sales zones, or identify product categories driving margin losses.

To address these challenges, there is a clear operational requirement for an end-to-end analytical pipeline. This pipeline must first clean, transform, and evaluate data programmatically through Python, and subsequently transform those refined data structures into dynamic, interactive executive dashboards using Microsoft Power BI for real-time decision support.

4.4 Project Objectives

The primary objective of this internship project was to design, develop, and deploy a comprehensive, multi-phase Data Analytics framework. This framework bridges programmatic data manipulation with interactive business intelligence reporting to resolve transactional visibility issues within retail operations.

The specific analytical and technical objectives established for the project include:

- **Data Hygiene & Structural Validation:** Implement automated data cleaning and pre-processing routines using Python (Pandas, NumPy) to handle missing values, resolve data type mismatches, remove duplicate transactional entries, and structure raw data for downstream consumption.
- **Exploratory Visual Analysis (EDA):** Perform in-depth programmatic exploratory data analysis to compute statistical summaries, uncover distributions, evaluate correlation patterns, and identify regional or categorical anomalies using visualization libraries such as Matplotlib and Seaborn.
- **Relational Data Modeling & ETL Integration:** Transform raw transactional tables in Power Query into an optimized dimensional model (Star Schema), establishing clear relationships between facts and dimensions while ensuring optimal data integrity and query execution speed.
- **Advanced KPI Computation via DAX:** Engineer custom Data Analysis Expressions (DAX) measures and calculated columns in Microsoft Power BI to dynamically derive critical business performance metrics, including Total Sales, Total Profit, Profit Margin %, Average Order Value (AOV), and Quantity Sold.
- **Interactive Executive Dashboard Design:** Build dynamic, user-centric visual interfaces containing interactive slicers, drill-down capabilities, geospatial sales maps, cross-filtering, and trend line analyses that empower stakeholders to evaluate regional, customer, and product performance in real time.
- **Actionable Business Strategy Formulation:** Translate synthesized analytical outputs and dashboard insights into strategic recommendations regarding inventory allocation, regional shipping optimization, discount structures, and product catalog focus areas.

4.5 Scope of the Project

The scope of this project encompasses the end-to-end design, implementation, and deployment of a modular data analytics pipeline using sample retail transactional data from an enterprise superstore environment. The boundaries of the project cover both programmatic exploratory analysis and interactive dashboard development.

Specifically, the project scope includes:

- **Data Ingestion & Pre-Processing Scope:**
 - Extraction and loading of multi-attribute retail datasets containing order details, shipping information, customer demographics, product categories, and financial metrics (sales, quantity, discount, profit).
 - Data cleaning routines involving missing value imputation, duplicate removal, date-string formatting, and schema standardization using Python.
- **Exploratory Data Analysis (EDA) Scope:**
 - Statistical profiling and univariate/bivariate feature analysis.
 - Generation of programmatic visualization plots (bar graphs, line charts, heatmaps, box plots) to evaluate sales variance, discount impacts, and regional distributions.
- **Business Intelligence & Data Modeling Scope:**
 - Building a consolidated Star Schema data model within Microsoft Power BI.
 - Creating custom DAX measures for real-time KPI aggregations, temporal sales comparisons (Year-over-Year, Month-over-Month), and margin evaluations.
- **Dashboard & Visual Reporting Scope:**
 - Designing multi-page interactive executive dashboards featuring dynamic filtering (by region, segment, year, ship mode, category), geospatial visualizations, and cross-highlighting features.
- **Out-of-Scope Elements:**
 - Live SQL database pipeline connections or real-time streaming data integration (analysis was conducted on static enterprise export files).
 - Predictive machine learning model deployment (e.g., automated demand forecasting or customer churn prediction algorithms), as the primary focus remained strictly on Descriptive and Diagnostic Analytics.

4.6 Technologies and Tools Used

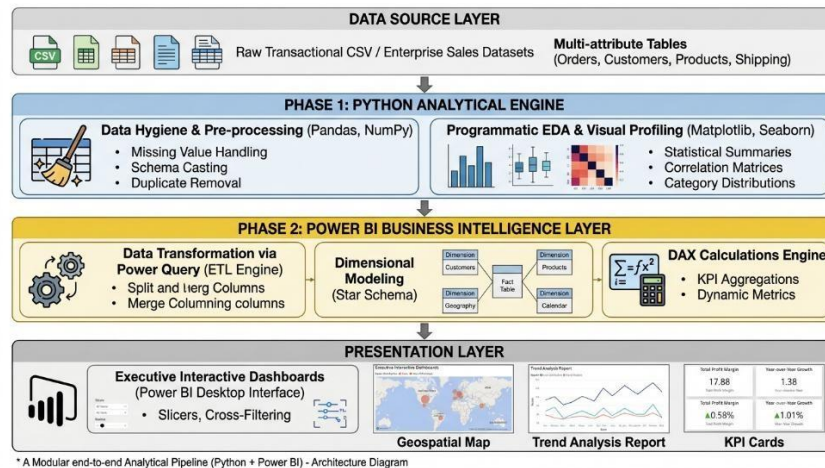
To execute the project from raw data ingestion to interactive dashboard deployment, a comprehensive suite of data engineering, analytical, and business intelligence software was utilized. The technology stack was strategically selected to leverage open-source Python libraries for exploratory data manipulation alongside industry-standard software for executive reporting.

The technologies and tools used throughout the project include:

- **Programming Language & Core Environment:**
 - **Python:** Used as the core programmatic language for scripting data cleaning, transformation pipelines, and initial exploratory data analysis.
 - **Jupyter Notebook / Anaconda:** Served as the interactive development environment (IDE) for executing step-by-step Python scripts, documenting analysis, and verifying visual outputs.
- **Python Data Processing & Analytical Libraries:**
 - **Pandas:** Employed for efficient structured data manipulation, tabular data frame processing, dataset merging, missing value imputation, and filtering operations.
 - **NumPy:** Leveraged for numerical computing, vector operations, and matrix calculations required for calculating analytical metrics.
- **Python Visual Data Exploration Libraries:**
 - **Matplotlib:** Utilized for building fundamental static visualization graphics, custom plot layouts, and statistical distribution charts.
 - **Seaborn:** Used for advanced statistical data visualization, including correlation heatmaps, categorical box plots, and multi-variable relationship distributions.
- **Business Intelligence & Dashboarding:**
 - **Microsoft Power BI Desktop:** Served as the primary enterprise BI environment for data transformation, dimensional data modeling, DAX measure calculation, and interactive visual interface engineering.
 - **Power Query Engine:** Utilized within Power BI for ETL (Extract, Transform, Load) operations, attribute split-column functions, data type assignments, and conditional column creation.
 - **Data Analysis Expressions (DAX):** Used to construct specialized functional expressions, dynamic KPI aggregations, and temporal time-intelligence metrics.
- **Data Formats & Storage:**
 - **CSV / Excel Data Files:** Structured relational transactional datasets containing order history, customer segmentation, product catalogs, shipping details, and financial parameters.

4.7 System Architecture

The architecture of the implemented Data Analytics solution follows a modular, end-to-end analytical pipeline. It converts raw, unstructured transactional records into refined operational models, ending with interactive executive dashboards. The pipeline is designed around a three-tier architecture: Data Ingestion & ETL, Exploratory Processing, and BI Presentation Layer.



Core Components of the Architecture

- **Data Ingestion Tier:** Acts as the entry point where raw, multi-column transactional CSV files are imported. These datasets contain raw attributes such as Order IDs, Order Dates, Customer Segments, Product Categories, Sales, Quantity, Discounts, and Profit Margins.
- **Exploratory & Transformation Tier (Python Engine):** Executes programmatic data cleaning and validation scripts. Pandas and NumPy standardize column formats, address null values, and remove duplicate entries. Matplotlib and Seaborn generate statistical plots and heatmaps to reveal baseline distributions and relationships between features before enterprise modeling begins.
- **Data Modeling & Analytics Engine (Power BI & DAX):** Operates inside Microsoft Power BI. Power Query performs structural ETL transformations, splitting raw flat tables into normalized Fact (sales transactions) and Dimension (Customers, Products, Geography, Calendar) tables to establish a Star Schema model. Custom DAX measures compile real-time, dynamic aggregations for KPIs like Total Profit Margin, Year-over-Year Growth, and Average Order Value.
- **Presentation Tier:** The final layer where analytical output is exposed to decision-makers. It houses multi-page, responsive dashboard screens configured with interactive slicers, geospatial maps, bar charts, and KPI cards, enabling real-time cross-filtering and drill-down analysis.

4.8 Methodology

The methodology adopted for this project follows a structured, sequential, and project-based approach, mirroring real-world data analytics workflows. By dividing the project into two distinct yet connected phases—programmatic preprocessing and interactive business intelligence—the methodology ensured that the final outputs were derived from clean, validated, and robust data foundations.

The end-to-end operational workflow is detailed below:

Phase 1: Exploratory Data Analysis (EDA) & Programmatic Visualization

This initial phase focused on leveraging the Python programming ecosystem to understand the dataset's structure, hygiene, and underlying statistical properties before attempting enterprise modeling.

1. **Requirement Gathering & Data Ingestion:** The process began by identifying the operational data needed to analyze the commercial superstore's performance. The raw transactional records, contained in flat CSV/Excel files, were loaded into the Jupyter Notebook environment using the Pandas library for initial ingestion and data frame creation.
2. **Data Cleaning & Pre-processing (ETL):** Raw data is rarely perfect. This critical step involved scripting automated routines to address significant data hygiene issues:
 - **Handling Null Values:** Identification of missing entries within critical columns (e.g., Customer Name, Sales, Profit) and applying appropriate imputation strategies or removing incomplete records where necessary.
 - **Duplicate Removal:** Scripts were executed to identify and purge duplicate transactional entries based on unique Order IDs to prevent revenue overestimation.
 - **Type Casting:** Attributes that were incorrectly formatted (e.g., monetary values as strings or dates as generic objects) were programmatically converted into proper numerical and datetime schemas (e.g., datetime64, float64).
3. **Visual Exploratory Analysis:** After data cleaning, statistical visual profiling was performed using Matplotlib and Seaborn. Programmatic scripts generated univariate plots to observe single-variable distributions, bivariate analysis to explore relationships between variables (e.g., Sales vs. Profit), and correlation heatmaps to visualize dependencies between different transactional features. This phase was essential for identifying baseline trends and anomalies programmatically.

Phase 2: End-to-End Business Intelligence & Dashboarding

This phase transitioned the analytical outputs from Phase 1 into Microsoft Power BI to build an interactive, executive-grade business intelligence reporting system.

1. **ETL & Multi-File Integration via Power Query:** The clean datasets from the programmatic phase were ingested into Power BI. While Python handles the main cleaning, the Power Query engine was utilized for structural transformations, such as splitting complex columns (e.g., Address columns), trimming whitespace, adding conditional columns, and merging lookup tables to optimize the dataset for reporting.
2. **Dimensional Modeling (Star Schema):** To ensure high-performance querying and accurate aggregations, a relational data model was designed. The transactional data was structured into a **Star Schema**, creating a primary **Fact Table** (containing sales transactions and numerical metrics) linked to several **Dimension Tables** (describing Customers, Products, Regions, and Time). Relationships (one-to-many) were established to enable dynamic filtering across different visual components.
3. **Data Analysis Expressions (DAX) Implementation:** Standard aggregations (sum, count) were insufficient for complex analysis. Advanced DAX measures were engineered to dynamically calculate sophisticated Key Performance Indicators (KPIs), such as **Profit Margin %**, **Average Order Value**, **Year-over-Year (YoY) Growth**, and **Shipping Lag Time**, allowing metrics to adjust instantly based on user-selected filters.
4. **UI/UX Dashboard Engineering:** The final step involved designing the interactive front-end. Clear visual hierarchies were created across multiple report pages (e.g., Executive Overview, Regional Performance, Customer Segmentation). Interactive elements, including slicers (for Date, Region, Category), drill-down capabilities (Product Category → Sub-Category), geospatial maps, and cross-filtering visual components, were deployed to empower stakeholders with self-service analytics.

4.9 Expected Outcomes

The completion of this Data Analytics project yields a set of structured, high-value outcomes that directly address the operational challenges of handling raw retail transactional data. By combining programmatic data preparation in Python with executive business intelligence in Microsoft Power BI, the project delivers both technical artifacts and actionable business value.

The core expected outcomes of the project include:

- **Automated Data Cleaning & Hygiene Pipelines:** A reusable Python framework capable of ingesting raw, unstructured transactional datasets and performing automated cleaning, null value imputation, duplicate removal, and schema validation, drastically reducing manual data preparation time for future reports.
- **In-Depth Exploratory Visual Analytics:** A comprehensive set of statistical visualizations and correlation heatmaps that document baseline sales distributions, identify loss-making product categories, and reveal historical buying patterns across customer segments.
- **Optimized Dimensional Data Model:** A high-performance Star Schema data model constructed within Power BI, establishing structured relationships between sales transactions (Fact) and dimensional lookups (Customers, Geography, Products, Calendar) to enable fast query responses and seamless data cross-filtering.
- **Dynamic Metric Computation via Advanced DAX:** A robust library of custom DAX measures that calculate real-time business Key Performance Indicators (KPIs)—including Total Profit Margins, Average Order Values, Shipping Lag Times, and Year-over-Year (YoY) Sales Growth—ensuring key metrics recalculate dynamically based on active filters.
- **Interactive Executive-Grade Dashboard:** A fully deployed, multi-page interactive dashboard engineered in Power BI Desktop featuring dynamic slicers, geospatial sales mapping, drill-down hierarchies, and visual cross-highlighting to provide executive stakeholders with self-service analytics capabilities.
- **Strategic Business Insights:** Actionable, data-backed operational recommendations regarding inventory allocation, regional discount optimization, high-value customer group target marketing, and shipping method refinements to improve overall profit margins.

4.10 Internship Proof



INTERNSHIP OFFER LETTER

Dear Ayush Chaudhary,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Data Analytics Intern
- Location & Mode: Pan India, Hybrid
- Duration : 08/07/2026 to 07/08/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.

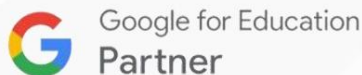
Best regards,

Sincerely,

Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 08/07/2026



Dr. Anurag Shrivastava
Talent Acquisition Manager



www.codectechnologies.in



support@codectechnologies.in



Chandivali IT hub, Mumbai

Chapter 5: Bibliography and References

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